

Total No. of Questions : 8]

SEAT No. :

PE4262

[Total No. of Pages : 2

[6582]-34

S.E. (Electronics & Computer Engineering)

COMPUTER ORGANIZATION

(2019 Pattern). (Semester-III) (204203)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Figures to the right indicates full marks.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Assume suitable data, if necessary.

- Q1)** a) Compare memory mapped I/O with I/O mapped I/O. [5]
- b) Explain in detail Direct Memory Access (DMA). [6]
- c) What are the evolutionary steps of I/O channel? Explain types of I/O channel? [6]

OR

- Q2)** a) Differentiate between programmed I/O and Interrupt driven I/O. [5]
- b) What is the use of DMA? Explain cycle stealing in DMA? [6]
- c) Discuss the following I/O mechanisms for transferring data with a neat flowchart: Programmed I/O and Interrupt Driven I/O. [6]

- Q3)** a) Explain different addressing modes with example. Discuss drawback of relative addressing mode. [5]
- b) Explain the following addressing modes with one example such. [6]
- i) Immediate
 - ii) Register Indirect
 - iii) Direct
- c) What is displacement addressing? Explain its types with calculation of effective address. [6]

OR

P.T.O.

- Q4)** a) What is machine instruction? Explain any three types of operations. [5]
b) List the features of 8086 microprocessors. [6]
c) Explain the following addressing modes along with suitable example:[6]
i) Direct addressing
ii) Indirect addressing
iii) Displacement addressing mode

- Q5)** a) Explain with example working of multistage pipeline. [5]
b) What are various hazards in instruction pipelining? Explain with example. [6]
c) Explain the instruction cycle in detail. [7]

OR

- Q6)** a) List and explain various ways in which an instruction pipeline can deal with conditional branch instructions. [5]
b) What is register organization? What are different types of registers? Explain in detail. [6]
c) Write a short note on superscalar execution and superscalar implementation. [7]
- Q7)** a) Write control sequence for execution of the instruction Add (R3), R1. [5]
b) Explain the design of multiplier control unit using Delay Element method. [6]
c) What is microprogrammed control unit design? Draw and explain basic structure of micro-programmed control unit. [7]

OR

- Q8)** a) Compare hardwired control and microprogrammed control unit. [5]
b) Write a control sequence for the following instruction for single bus organization: SUB (R3), R1. [6]
c) Explain in detail microinstruction sequencing organization. [7]

